INCH-POUND

MIL-F-48425A (AR)
AMENDMENT 3
31 July 1989

SUPERSEDING
AMENDMENT 2
9 May 1983

MILITARY SPECIFICATION

FIN ASSEMBLY, M24

This Amendment forms a part of Military Specification MIL-F-48425A (AR), dated 11 September 1980, and is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.2, Delete: "ASTM E-18, Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials."

PAGE 3

3.4, Delete: "fin housing" where it appears (2 places), and substitute "cartridge housing."

PAGE 4

4.3.1: Add the following to the list of components submitted for first article testing:

Part Description	Drawings	Quantity
Flat tensile specimen	See 4.5.3.2	2
Round tensile specimen	See 4.5.3.2	2
Flat fully prepared microstructure specimen	See 4.5.3.3.1	2
Round fully prepared microstructure specimen	See 4.5.3.3.2	2
Macroetch specimens from 1 extruded bar of housing stock	See 4.5.2	2
Macroetch specimens from 1 extruded bar of fin stock	See 4.5.2	2
Blade security specimens	See 4.5.6	5

Last 2 paragraphs: Delete in their entirety and substitute:

"Macroetch specimens shall be identified as to their origin (front or rear of billet)."

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PAGE 5

Table I, First Article Inspection:

Delete "Hydrotest, to destruction, 10 (a), 3.2, 4.5.4."

Delete "Extrusion Blanks (housing)," and substitute, "Fully heat-treated cartridge housing stock."

Delete "Hardness test 2(c), 3.1, 4.5.3.1."

Microstructure examination: Delete "1, 3.3, 4.5.3.3," and substitute "2(C), 3.3, 4.5.3.3.2."

Delete "Extrusion Blanks (Fin)" and substitute "Fully heat-treated fin stock."

Add: "Microstructure examination, 2, 3.3, 4.5.3.3.1."

Note a: Delete in its entirety.

Note c: Delete in its entirety and substitute, "(C) Microstructure specimens shall be taken from either end of each round tensile specimen submitted for testing."

Note d: Delete in its entirety and substitute "(d) Largest standard flat-type specimens available."

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* 4.4.1.1 Delete in its entirety and substitute:

"4.4.1.1 Cartridge housing lot. A cartridge housing lot shall consist of blanks from the same heat treatment lot of the same alloy that have been heat treated in the same heat treating equipment at the same temperature in one unchanged process. A heat treatment lot may consist of more than one solution heat treatment batch, however, the samples for tensile and microstructure tests hereinafter prescribed must be selected from and represent each solution heat treatment batch. A solution heat treatment batch shall consist of one lot of the same alloy that has been thermally treated by the same method in the same furnace equipment at the same time and temperature in one unchanged process. The identification and segregation of each solution heat treatment batch is not required after acceptable compliance with the tensile, and microstructure test."

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* 4.4.1.2 Delete in its entirety and substitute:

"4.4.1.2 Fin stock lot. The extrusion lot shall consist of extruded stock of the same alloy that has been thermally treated by the same method in the same furnace equipment at the same temperature and time in one unchanged process regardless of the solution heat treating method utilized. Each extrusion lot representing material that has been solution heat treated after extrusion must be solution heat treated at the same time in the same furnace using the same temperature and time in the furnace. The extrusion lot thus constituted shall be the basis for selecting the sample blanks for the tension and microstructure tests hereinafter prescribed."

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* 4.4.2.b: Delete the following: "For inspection of para. 4.4.2.4 Minor 205 (Length of rear thread, max.): Inspect 1 housing at the end of each day's production by cutting the rear of the housing in half (longitudinally). Length shall be measured from the rear face to the end of the thread's run out. Any housing which does not meet the drawing requirement shall cause the rejection of all material produced since the last successful inspection."

PAGE 8

4.4.2.1: Delete "Fin and Housing Extrusion Bars" and substitute "Fin and Housing Extrusion bars prior to the final heat treatment."

PAGE 9

4.4.2.2: Delete "Fin and Housing Extrusion Blanks" and substitute "Fully heat-treated Fin and Housing Stock."

Special Category: Delete "Special a, Hardness Test, 80, 3.1, 4.5.3.1."

Note a: Delete in its entirety and substitute "(a) Four specimens total (2 round from housing stock, 2 flat from fin stock). Failure of any sample shall cause rejection of the applicable solution heat treatment batch or press-quench lot. See para. 4.4.1."

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Note b: Delete in its entirety and substitute "(b) Four specimens total (2 round, 2 flat). Failure of any sample shall cause rejection of the applicable solution heat treatment batch or press-quench lot. See para. 4.4.1."

PAGE 11

* Add majors:

119	True position of cartridge cavity with pitch diameter of rear thread min. length	.40%	3.2 Gage
120	Pitch diameter of forward thread, max.	.40%	3.2 Gage
121	Pitch diameter of forward thread, min.	.40%	3.2 Gage
122	Pitch diameter of rear thread, max.	.40%	3.2 Gage
123	Pitch diameter of rear thread, min.	.40%	3.2 Gage
124	Minor diameter of rear thread, max.	.40%	3.2 Gage
125	Diameter of cartridge cavity.	.40%	3.2 Gage

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4.4.2.4, Housing

Major 116: Delete "4.5.5.2" and substitute "Visual" in Inspection Method column.

* Minor 205: Delete "Tool control/4.4.2.b" and substitute "Gage" in Inspection Method column.

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* 4.4.2.6, Fin assembly: Add the following:

"Major 115, Blade security test, .025%, 3.2, Gage/4.5.6."

* Delete Majors 102, 104, 105, 106, 107, 108, and 109 in their entirety. Delete Note "a" in its entirety.

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4.5.3.1, Hardness: Delete in its entirety.

4.5.3.1.1: Delete in its entirety.

* 4.5.3.2 Delete in its entirety and substitute:

"4.5.3.2 Tension test. Following final heat treatment, one specimen shall be submitted per 1,000 pounds of stock from both housing stock and fin stock. The fin specimens shall be the largest standard flat-type specimens obtainable from the blade area and the housing specimens shall be the largest standard round-type specimens obtainable from housing stock. Specimens shall be taken longitudinally. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the applicable solution heat treatment batch or press-quench lot. The test criteria for all tensile specimens shall be as prescribed by ASTM Method E-8. The number of specimens and their location is to verify compliance with the mechanical properties requirements. The Government shall not be restricted from testing additional specimens at any location to determine that the mechanical properties meet the drawing requirements for yield strength and elongation."

4.5.3.2.1: Delete paragraph in its entirety.

4.5.3.2.2: Delete paragraph in its entirety.

PAGE 20

4.5.3.3, Microstructure examination: Delete paragraph in its entirety and add the following:

"4.5.3.3 Microstructure examination.

4.5.3.3.1 Fin stock. From the fin blade adjacent to each of the flat samples submitted for tension testing, one wafer approximately 1/2 inch square shall be prepared for metallographic examination from undeformed material. When suitably prepared, the specimens shall be etched with Keller's reagent and examined by scanning under a metallurgical microscope at a minimum magnification of 500 diameters. Overheating shall be evidenced by the presence of rosettes

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in the microstructure or incipient melting at the grain boundaries (see 6.5). Evidence of the above shall cause rejection of the applicable solution heat treat batch or press-quench lot.

4.5.3.3.2 Housing stock. Metallographic specimens from housing material shall be cut from a transverse section at either end of each round specimen submitted for tension testing. Preparation and examination for overheating, porosity etc. shall be as described in 4.5.3.3.1."

* 4.5.3.4 Delete in its entirety.

4.5.4 Hydrostatic test

line 10: Delete the following sentence: "The section where the fin is assembled to the housing shall be fully supported on the outside diameter during test."

line 24: Delete the following sentence: "For first article inspection only, the test pressure shall be increased until the housing ruptures and the maximum pressure shall be recorded for information purposes."

PAGE 21

4.5.5.2: Delete paragraph in its entirety and substitute:

"4.5.5.2 Excessive flash holes. The sample fin housings shall be visually inspected for the presence of excess flash holes."

* 4.5.6 Blade security test add the following:

"4.5.6 Blade security test: The housing portion shall be held securely in a fixture while 375 ± 25 inch-pounds torque is applied for 3 seconds to the blade portion of the assembly. Rotation of the blades around any housing shall be cause for rejection. If the blade cants are contacted during testing, the assembly must be discarded whether or not the assembly passes."

* 6.3, line 4: Delete: DRDAR-QAR-I and substitute AMSMC-QAT-I.

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* 6.8 Add the following: Direct submission to Commander, U.S. Army Armament Research and Development Command, ATTN: AMSMC-QAT-B, Picatinny Arsenal, N.J. 07806-5000.

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